PYTHON LAB RECORD

10 Programs with Aim, Procedure, Flowchart, Program and Sample Output

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Prepared from the 10 Python files uploaded in this conversation.

Each program is presented with its original source code and a sample run. Sample inputs are provided only to demonstrate the output.

Program 1: Progressive Electricity Billing

Aim

To calculate an electricity bill using unit-based slabs, add the fixed charge, and compare it with the previous month's bill to identify an unusual increase.

Procedure

1. Read electricity units.

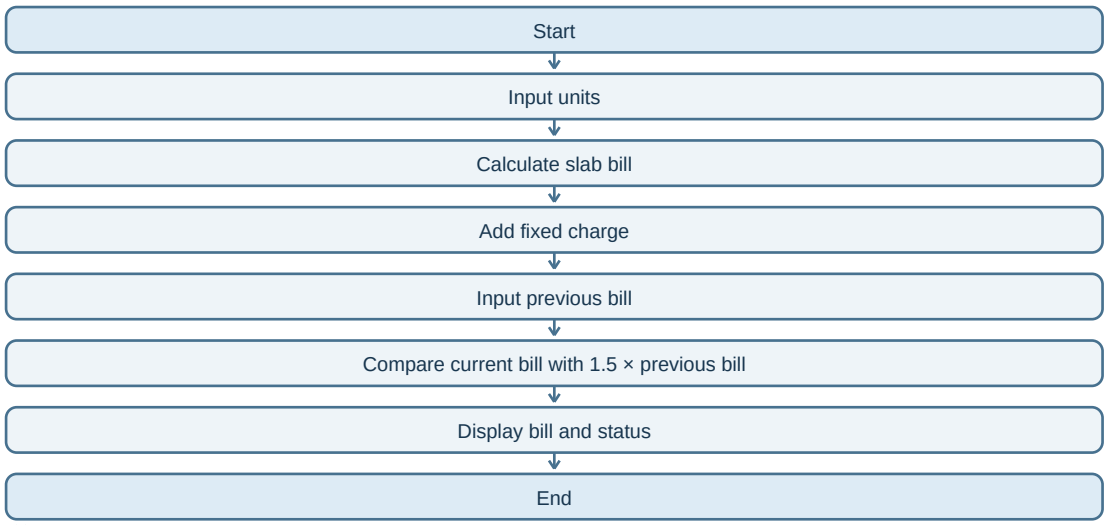
2. Calculate the energy charge using the applicable slab.

3. Add the fixed charge of 50.

4. Read the previous month's bill.

5. If the current bill is more than 1.5 times the previous bill, display an unusual-increase message; otherwise display a normal-bill message.

Flowchart



Program

```
units = int(input("Enter electricity units: "))

if units <= 100:
    bill = units * 2
elif units <= 200:
    bill = 200 + (units - 100) * 3
else:
    bill = 500 + (units - 200) * 5

fixed = 50
total = bill + fixed

print("Electricity Bill:", total)
month1 = int(input("Enter previous month bill: "))
month2 = total

if month2 > month1 * 1.5:
    print("Unusual increase in bill!")
else:
    print("Normal bill")
```

Sample Output

```
Enter electricity units: Electricity Bill: 400
Enter previous month bill: Unusual increase in bill!
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
```

```
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 2: University Scholarship Eligibility

Aim

To check whether a student is eligible for a scholarship based on aggregate marks, subject marks, attendance, and family income.

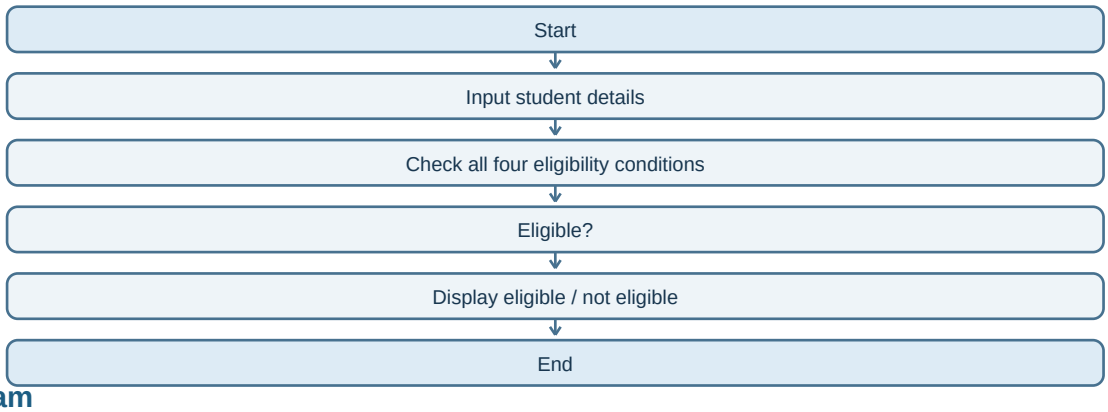
Procedure

1. Read the student's name, aggregate marks, subject marks, attendance, and family income.

2. Check whether marks are at least 75, subject marks are at least 60, attendance is at least 75, and income is not more than 200000.

3. Display scholarship eligibility based on the condition.

Flowchart



Program

```
name = input("Enter student name: ")
marks = int(input("Enter aggregate marks: "))
subject = int(input("Enter subject marks: "))
attendance = int(input("Enter attendance: "))
income = int(input("Enter family income: "))

if marks >= 75 and subject >= 60 and attendance >= 75 and income <= 200000:
    print("Scholarship Eligible")
else:
    print("Not Eligible")
```

Sample Output

```
Enter student name: Enter aggregate marks: Enter subject marks: Enter attendance: Enter family income: Scholarsh
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 3: Student Attendance Calculator

Aim

To calculate a student's attendance percentage and find how many additional consecutive classes are needed to reach 75% attendance.

Procedure

1. Read attended classes and total classes.

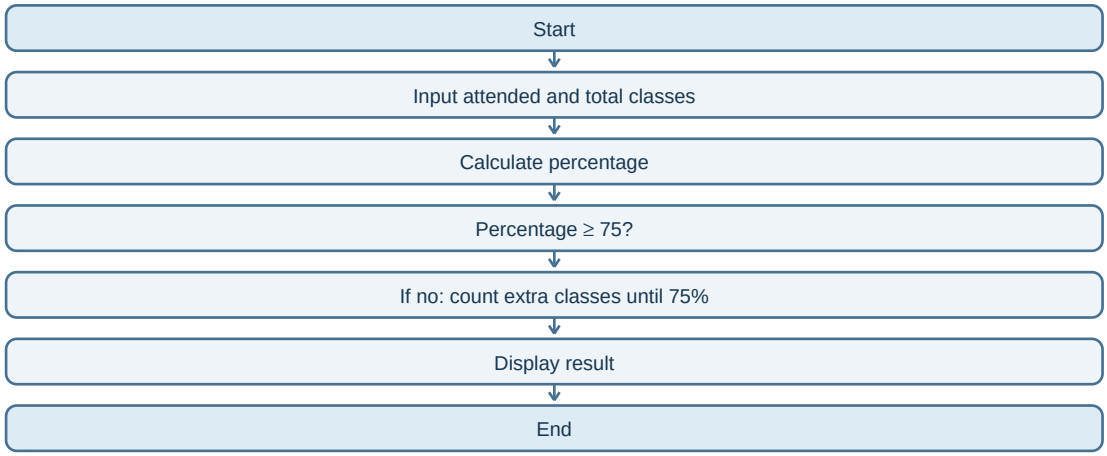
2. Calculate attendance percentage.

3. If it is at least 75%, display that the student is eligible.

4. Otherwise, increase the number of future attended classes one by one until the percentage reaches 75%.

5. Display the number of additional consecutive classes needed.

Flowchart



Program

```
attended = int(input("Enter attended classes: "))
total = int(input("Enter total classes: "))

percentage = (attended / total) * 100

print("Current Attendance:", percentage, "%")

required = 75

if percentage >= required:
    print("You are eligible")
else:
    extra = 0
    while ((attended + extra) / (total + extra)) * 100 < required:
        extra += 1

    print("You need", extra, "more consecutive classes")
    print("Required attendance achieved!")
```

Sample Output

```
Enter attended classes: Enter total classes: Current Attendance: 60.0 %
You need 60 more consecutive classes
Required attendance achieved!
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
```

```
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 4: Student Performance Analysis

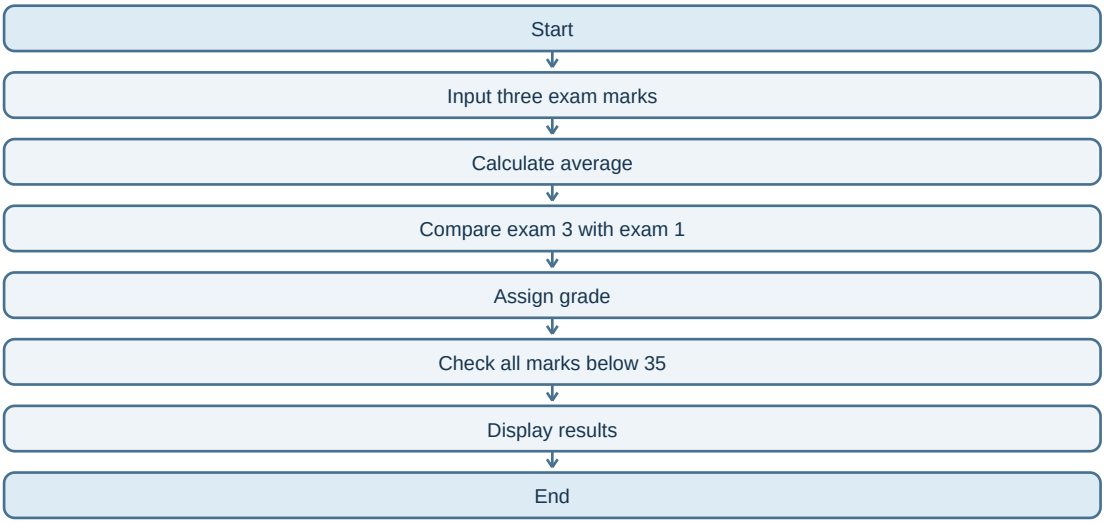
Aim

To calculate the average of three exam marks, compare performance, and display the grade.

Procedure

1. Read the student's name and marks in three exams.
2. Calculate and display the average.
3. Compare the third exam mark with the first to identify improvement, decline, or stable performance.
4. Assign Excellent, Average, or Poor grade based on the average.
5. Check whether all three marks are below 35.

Flowchart



Program

```
name = input("Enter student name: ")

m1 = int(input("Enter first exam marks: "))
m2 = int(input("Enter second exam marks: "))
m3 = int(input("Enter third exam marks: "))

average = (m1 + m2 + m3) / 3

print("Average Marks:", average)

if m3 > m1:
    print("Performance Improved")
elif m3 < m1:
    print("Performance Declined")
else:
    print("Performance Stable")

if average >= 75:
    print("Grade: Excellent")
elif average >= 50:
    print("Grade: Average")
else:
    print("Grade: Poor")

if m1 < 35 and m2 < 35 and m3 < 35:
    print("Consistently Weak Performance")
```

Sample Output

```
Enter student name: Enter first exam marks: Enter second exam marks: Enter third exam marks: Average Marks: 70.0
Performance Improved
Grade: Average
Program error:
```

```
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 5: Timetable Conflict Detection

Aim

To check whether two timetable entries have a time conflict or a room conflict.

Procedure

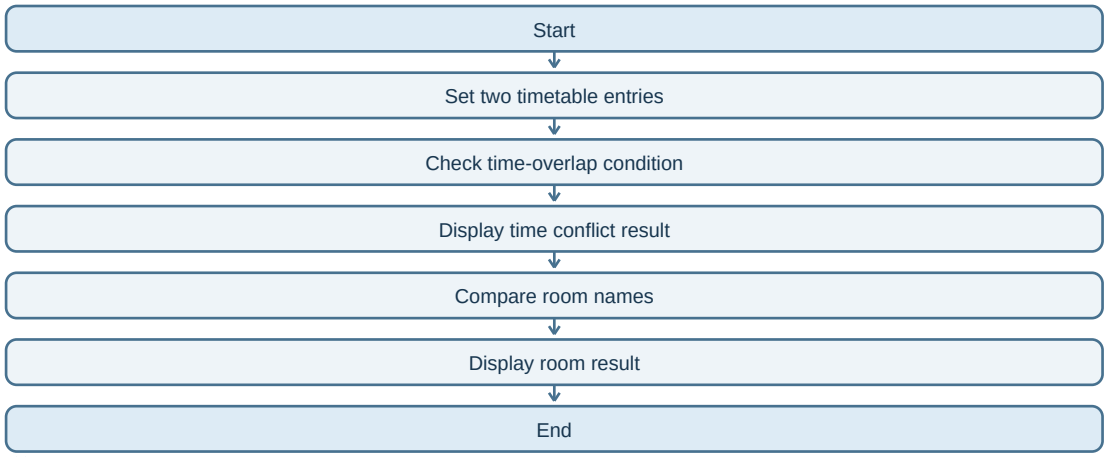
1. Create two timetable entries containing day, start time, end time, and room.

2. Display both timetable entries.

3. Check the program's time-overlap condition and display the result.

4. Compare the room names and display whether a room conflict exists.

Flowchart



Program

```
t1 = ("Monday", "9:00", "10:00", "Room A")
t2 = ("Monday", "9:30", "10:30", "Room B")

print("Timetable 1:", t1)
print("Timetable 2:", t2)

if t1[0] == t2[0] and t1[2] > t2[1] and t1[1] < t2[2]:
    print("Time Conflict Detected!")
else:
    print("No Time Conflict")

if t1[3] == t2[3]:
    print("Room Conflict Detected!")
else:
    print("Rooms are Available")
```

Sample Output

```
Timetable 1: ('Monday', '9:00', '10:00', 'Room A')
Timetable 2: ('Monday', '9:30', '10:30', 'Room B')
No Time Conflict
Rooms are Available
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 6: Student Marks Analysis

Aim

To calculate each student's total marks, find each subject's average, and identify the top student.

Procedure

1. Store the marks of three students in a two-dimensional list.

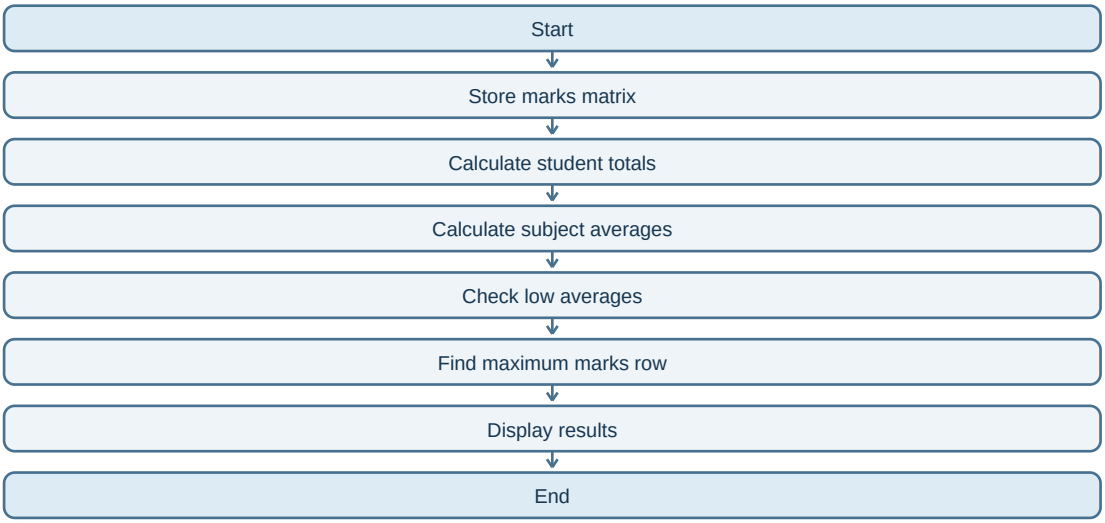
2. Calculate and display each student's total.

3. Calculate and display the average for each subject.

4. Display Low Average if a subject average is below 60.

5. Find and display the student number associated with the maximum marks row.

Flowchart



Program

```
marks = [
    [80, 70, 90],
    [60, 50, 65],
    [90, 85, 95]
]

for i in range(3):
    print("Student", i+1, "Total:", sum(marks[i]))

for j in range(3):
    avg = (marks[0][j] + marks[1][j] + marks[2][j]) / 3
    print("Subject", j+1, "Average:", avg)

    if avg < 60:
        print("Low Average")

print("Top Student:", marks.index(max(marks)) + 1)
```

Sample Output

```
Student 1 Total: 240
Student 2 Total: 175
Student 3 Total: 270
Subject 1 Average: 76.66666666666667
Subject 2 Average: 68.33333333333333
Subject 3 Average: 83.33333333333333
Top Student: 3
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
```

```
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 7: Team Ranking System

Aim

To rank teams using points and use scores as the tie-breaker.

Procedure

1. Store team names, points, and scores in lists.

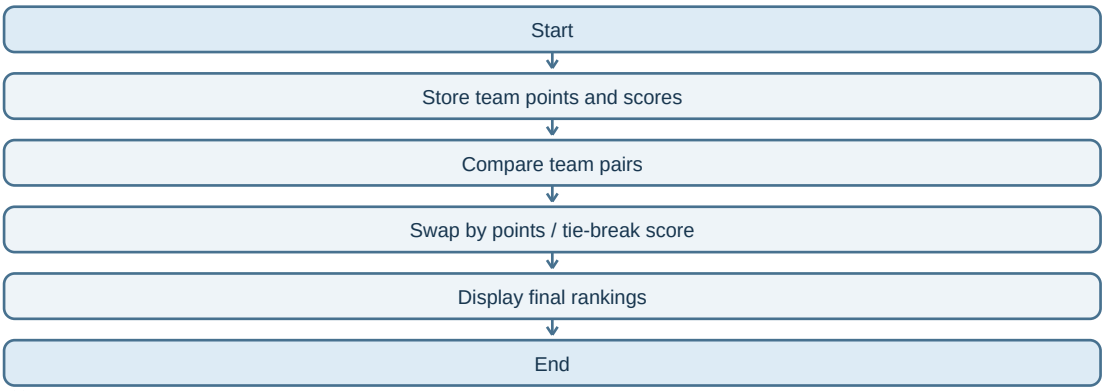
2. Display the original team details.

3. Compare each pair of teams.

4. Swap their positions when a team has more points, or equal points with a higher score.

5. Display the final ranking.

Flowchart



Program

```
teams = ["Team A", "Team B", "Team C"]
points = [6, 9, 6]
scores = [10, 15, 12]

for i in range(3):
    print(teams[i], "Points:", points[i],
          "Score:", scores[i])

for i in range(3):
    for j in range(i + 1, 3):
        if points[j] > points[i] or (points[j] == points[i] and scores[j] > scores[i]):
            points[i], points[j] = points[j], points[i]
            scores[i], scores[j] = scores[j], scores[i]
            teams[i], teams[j] = teams[j], teams[i]

print("\nFinal Rankings")

for i in range(3):
    print(i + 1, teams[i], points[i], "Points")
```

Sample Output

```
Team A Points: 6 Score: 10
Team B Points: 9 Score: 15
Team C Points: 6 Score: 12

Final Rankings
1 Team B 9 Points
2 Team C 6 Points
3 Team A 6 Points
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
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  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
```

```
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 8: Student Academic Risk Analysis

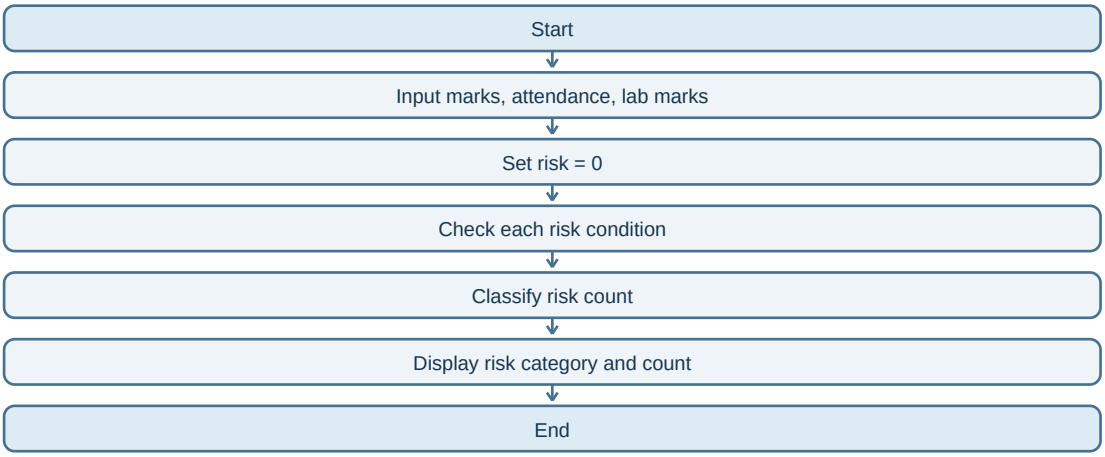
Aim

To identify student academic risk factors based on marks, attendance, and lab marks, and classify the overall risk.

Procedure

1. Read the student's name, marks, attendance, and lab marks.
2. Start the risk count at zero.
3. Increase the count for marks below 40, attendance below 75, and lab marks below 40.
4. Classify zero risks as No Risk, one risk as Medium Risk, and two or more risks as High Risk.
5. Display the total number of risk factors.

Flowchart



Program

```
name = input("Enter student name: ")
marks = int(input("Enter marks: "))
attendance = int(input("Enter attendance: "))
lab = int(input("Enter lab marks: "))

risk = 0

if marks < 40:
    print("Low Marks Risk")
    risk += 1

if attendance < 75:
    print("Low Attendance Risk")
    risk += 1

if lab < 40:
    print("Low Lab Marks Risk")
    risk += 1

if risk == 0:
    print("Category: No Risk")
elif risk == 1:
    print("Category: Medium Risk")
else:
    print("Category: High Risk")

print("Total Risk Factors:", risk)
```

Sample Output

```
Enter student name: Enter marks: Enter attendance: Enter lab marks: Low Marks Risk
Low Attendance Risk
Low Lab Marks Risk
Category: High Risk
Total Risk Factors: 3
Program error:
```

```
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
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  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 9: Scholarship Allocation

Aim

To select up to two eligible students based on marks, attendance, and family income, then rank them by marks.

Procedure

1. Read the number of students.

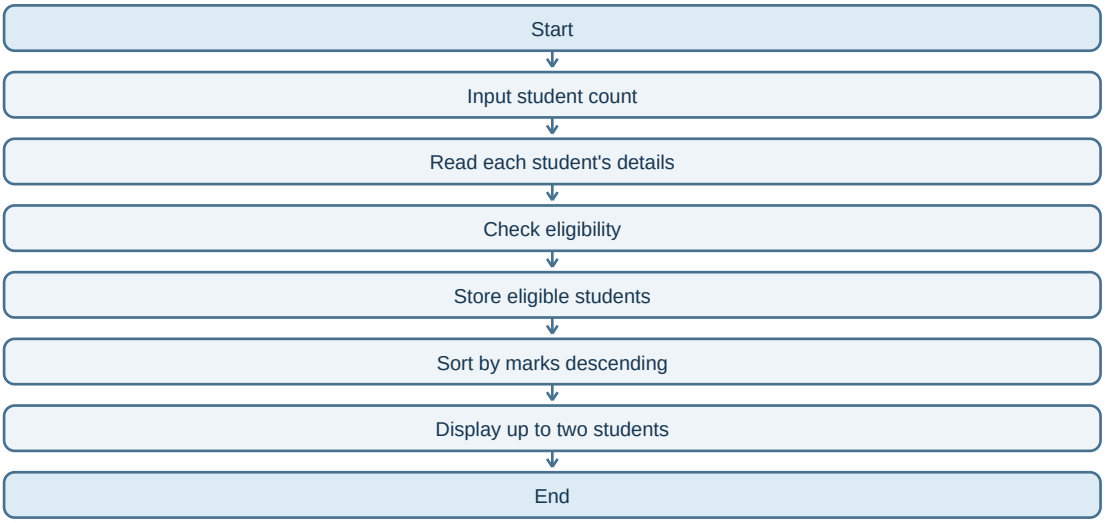
2. For each student, read name, marks, attendance, and family income.

3. Add students who have marks of at least 75, attendance of at least 75, and income not more than 200000 to the eligible list.

4. Sort eligible students by marks in descending order.

5. Display up to two selected students.

Flowchart



Program

```
students = []

n = int(input("Enter number of students: "))

for i in range(n):
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    attendance = int(input("Enter attendance: "))
    income = int(input("Enter family income: "))

    if marks >= 75 and attendance >= 75 and income <= 200000:
        students.append([name, marks])

students.sort(key=lambda x: x[1], reverse=True)

quota = 2

print("\nSelected Students:")

for i in range(min(quota, len(students))):
    print(students[i][0], students[i][1])
```

Sample Output

```
Enter number of students: Enter student name: Enter marks: Enter attendance: Enter family income: Enter student
Selected Students:
Sita 90
Ravi 80
Program error:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
```

```
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
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  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
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artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Program 10: Environmental Reading Analysis

Aim

To calculate the average, highest, and lowest environmental readings and classify the environment.

Procedure

1. Use the stored environmental readings.

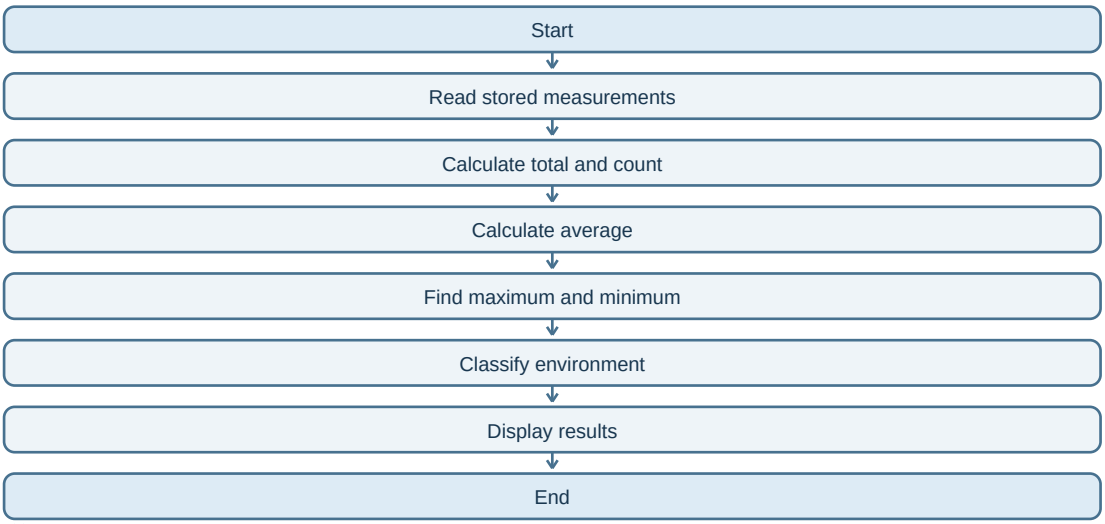
2. Visit every reading and calculate the total and count.

3. Calculate and display the average reading.

4. Find and display the highest and lowest readings.

5. If the average is at most 50, display Good; otherwise display Poor.

Flowchart



Program

```
readings = [
    [40, 50, 60],
    [70, 80, 90],
    [30, 40, 50]
]

total = 0
count = 0

for row in readings:
    for value in row:
        total += value
        count += 1

average = total / count

print("Average Reading:", average)
print("Highest Reading:", max(max(row) for row in readings))
print("Lowest Reading:", min(min(row) for row in readings))

if average <= 50:
    print("Environment: Good")
else:
    print("Environment: Poor")
```

Sample Output

```
Average Reading: 56.666666666666664
Highest Reading: 90
Lowest Reading: 30
Environment: Poor
Program error:
Spreadsheet runtime warmup failed during python startup
```

```
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_fea
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_dra
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```